

Sample End Semester Question Paper - 2025

Subject: Computer Networks | Code: CS402 | Semester: 4

Branch: CSE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. What is a subnet mask?
2. Define ARP.
3. What is the main difference between TCP and UDP?
4. Name two applications that use UDP.
5. What is DNS?
6. Name the seven layers of the OSI model.
7. What is a protocol?
8. State one difference between datagram and virtual circuit.
9. What is a frame?
10. State one use of a checksum.

PART B - Long questions (10 x 8 = 80 Marks)

1. Explain the operation of DNS, HTTP and FTP.
2. Compare OSI and TCP/IP models layer by layer.
3. Explain the OSI reference model with the function of each layer.
4. Explain data link layer functions: framing, error control and flow control.
5. Explain CRC error detection with a worked example.
6. Explain sliding window protocol with an example.
7. Explain classful IP addressing and subnetting with a worked example.
8. Explain ARP and ICMP operation with message sequences.
9. Explain TCP: three-way handshake, flow control and congestion control.
10. Explain routing: static, distance vector and link state with examples.